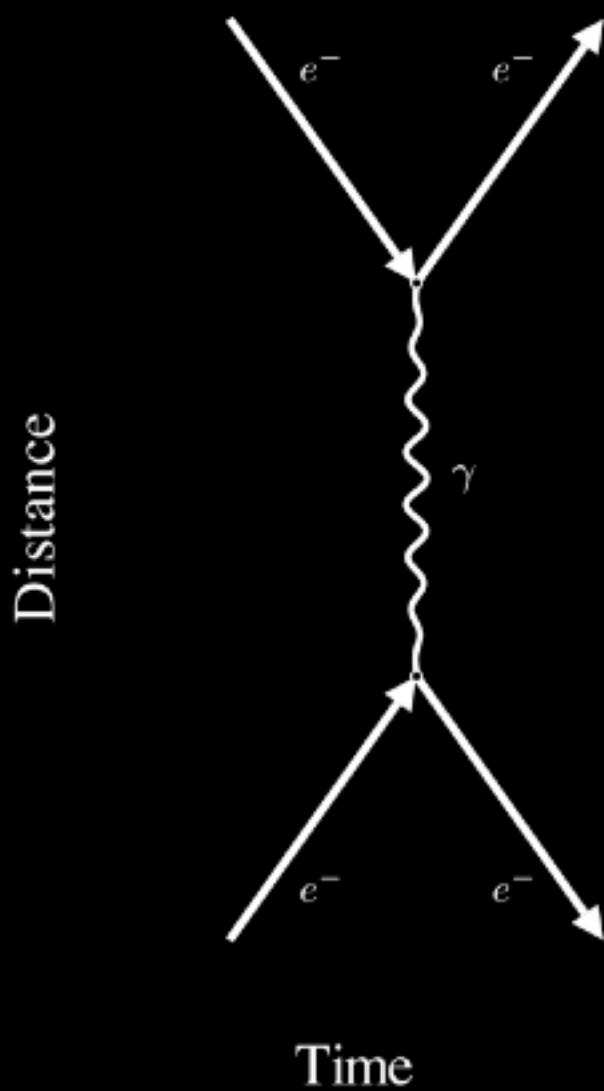


$$e^- e^- \rightarrow e^- e^-$$

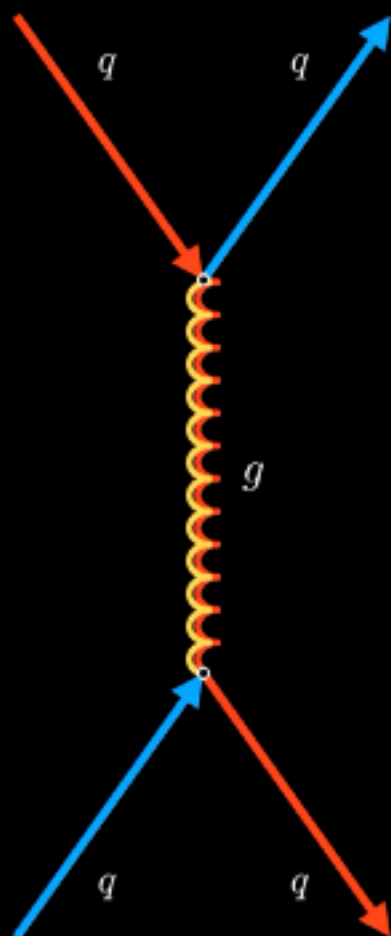
QED scattering with virtual photon exchange



$$q q \rightarrow q q$$

QCD scattering with virtual gluon exchange

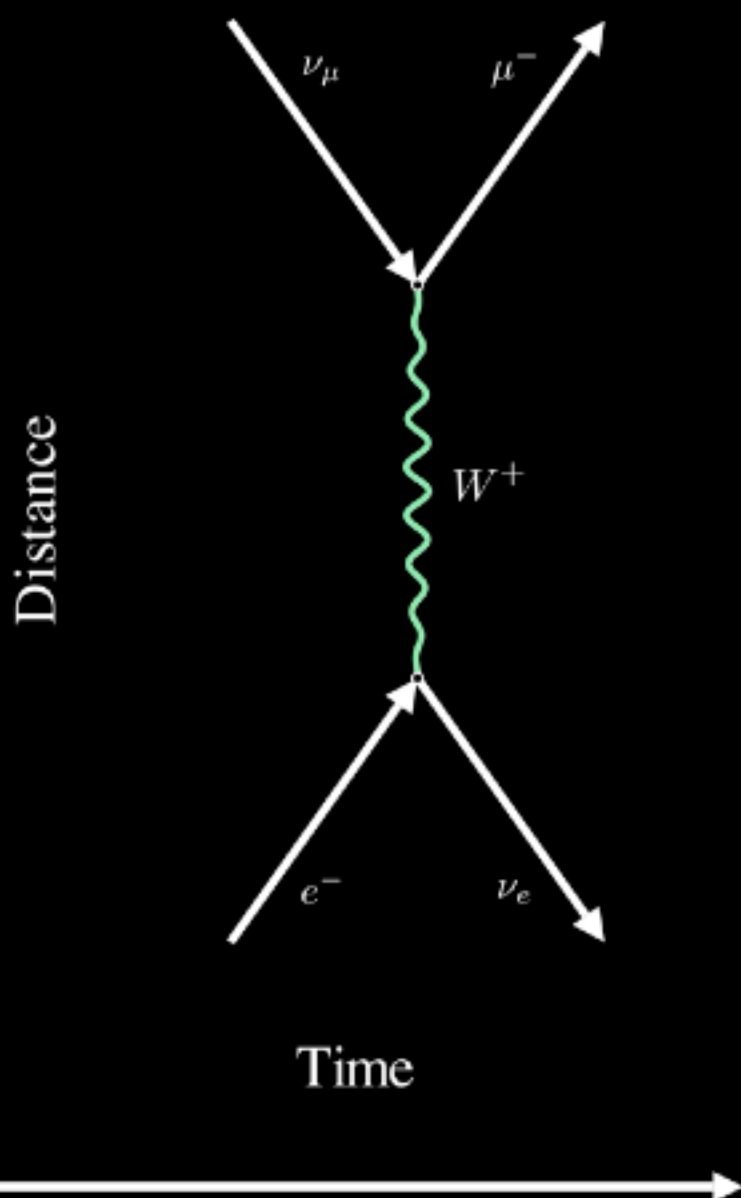
Distance



Time

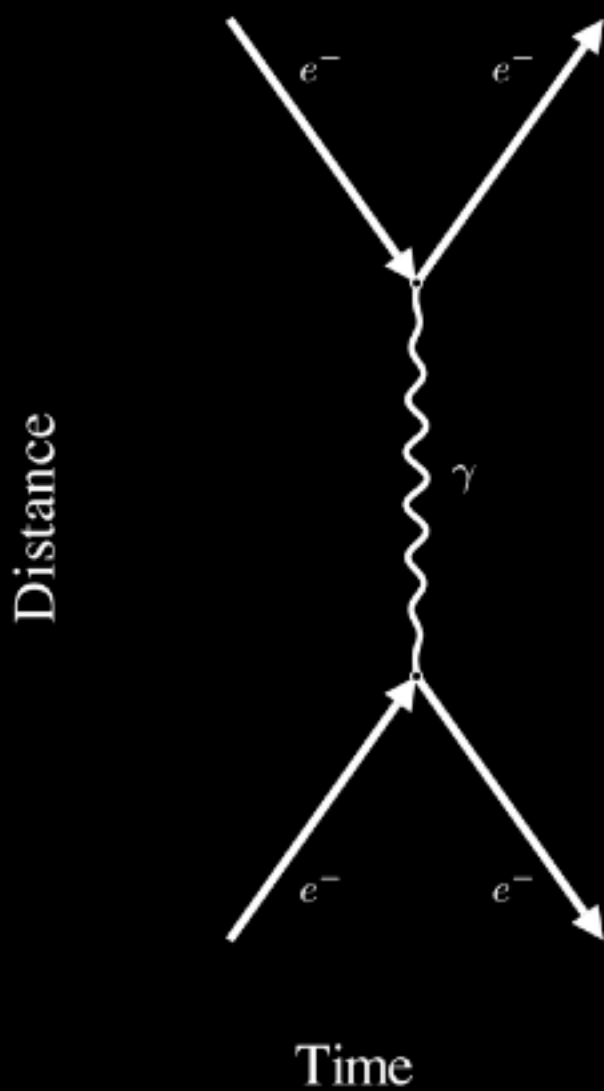
$$\nu_{\mu} e^{-} \rightarrow \nu_e \mu^{-}$$

Charged-current weak scattering with W exchange



$$e^- e^- \rightarrow e^- e^-$$

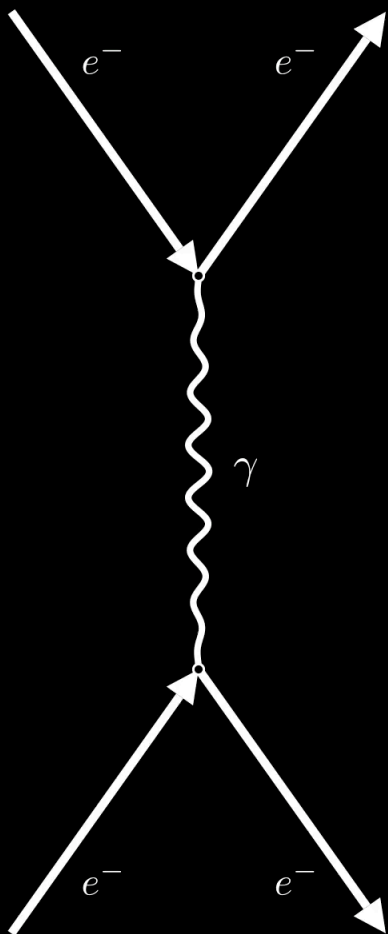
QED scattering with virtual photon exchange



$$e^- e^- \rightarrow e^- e^-$$

QED scattering with virtual photon exchange

Distance



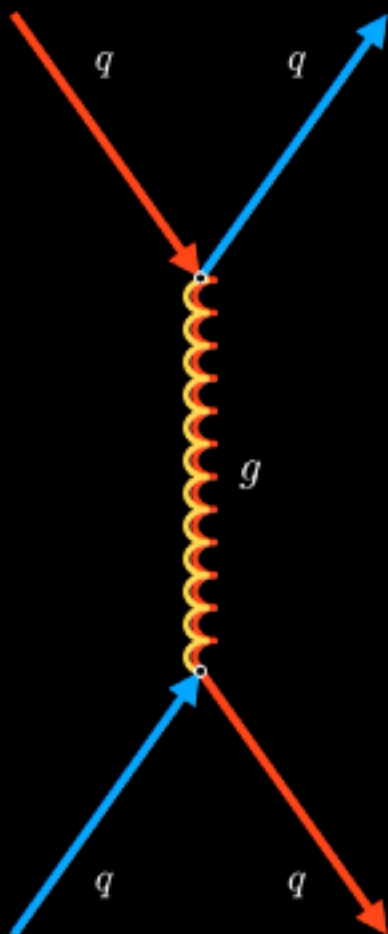
Time

$$q q \rightarrow q q$$

QCD scattering with virtual gluon exchange

Distance

Time

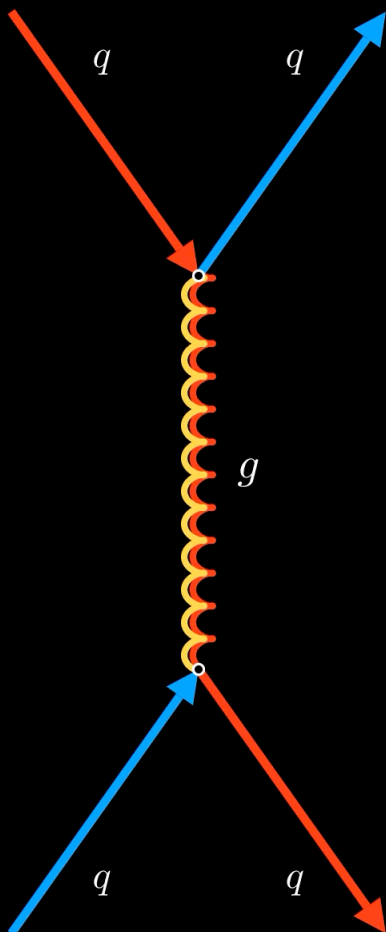


$$q q \rightarrow q q$$

QCD scattering with virtual gluon exchange

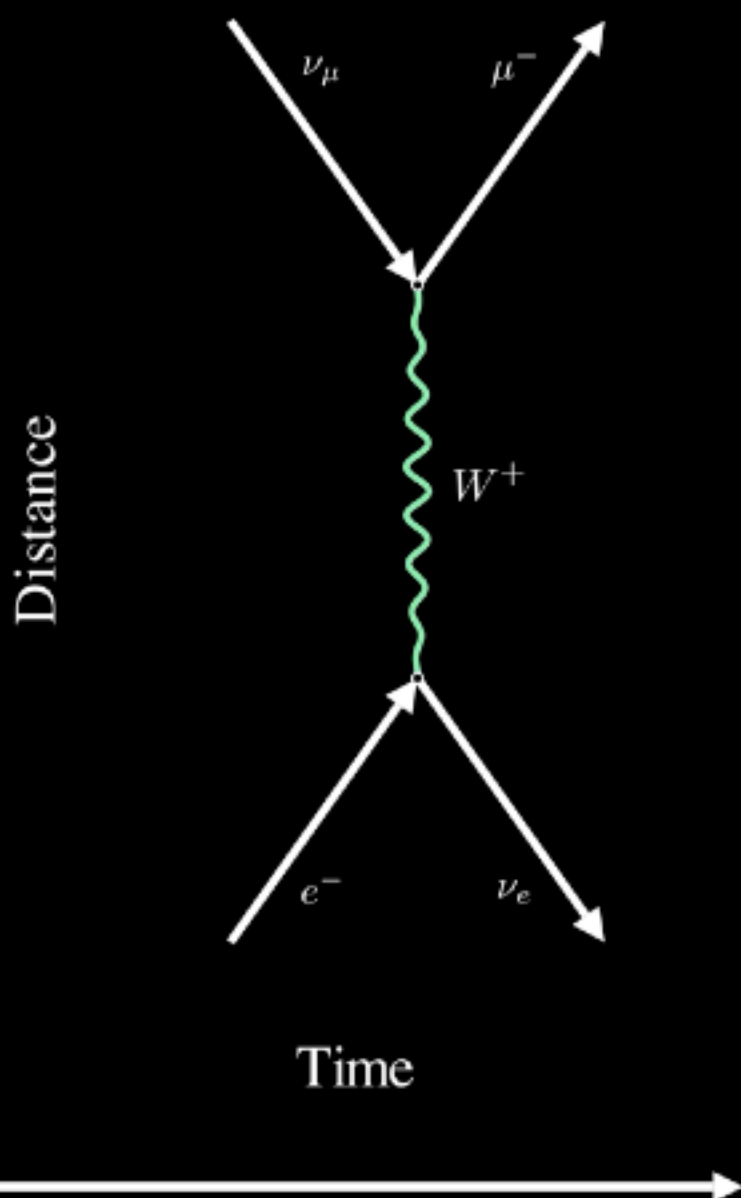
Distance

Time



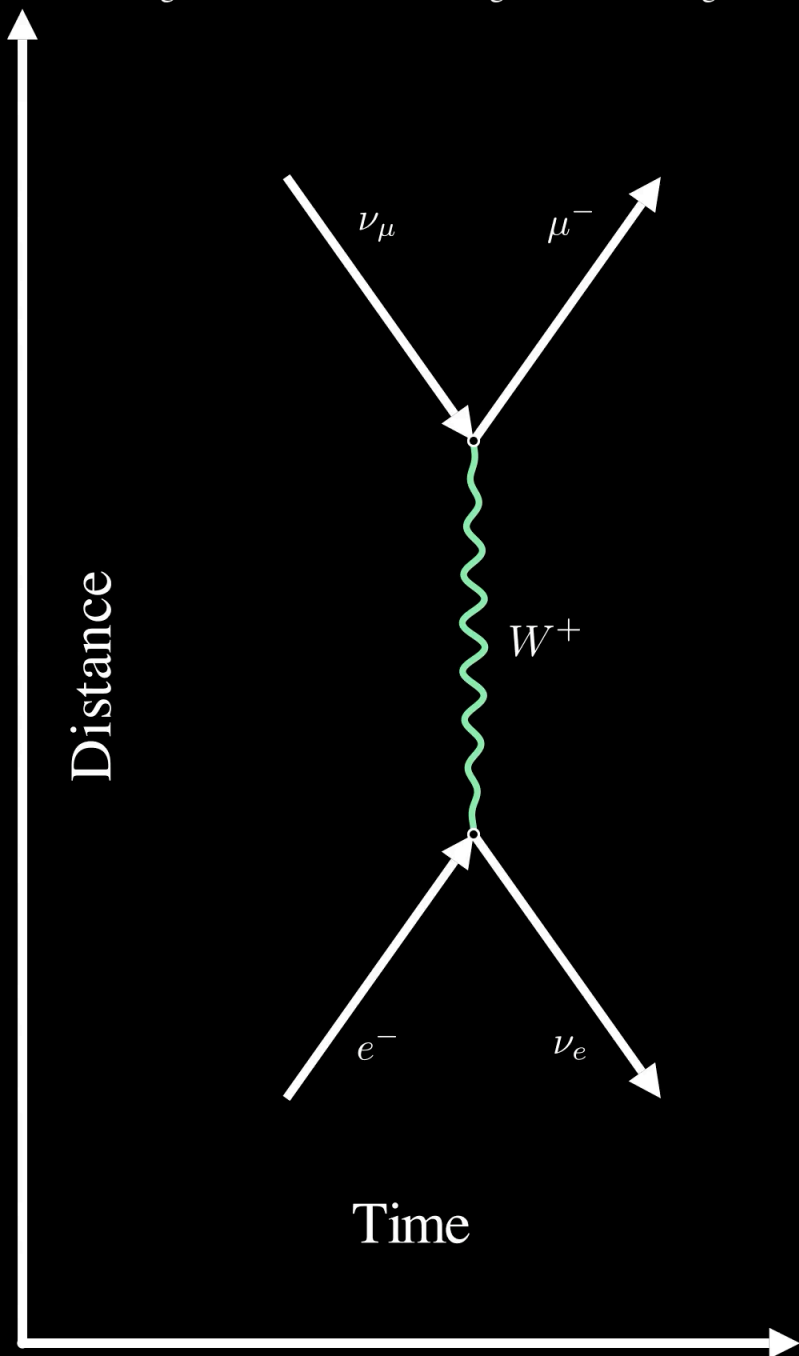
$$\nu_{\mu} e^{-} \rightarrow \nu_e \mu^{-}$$

Charged-current weak scattering with W exchange



$$\nu_{\mu} e^{-} \rightarrow \nu_e \mu^{-}$$

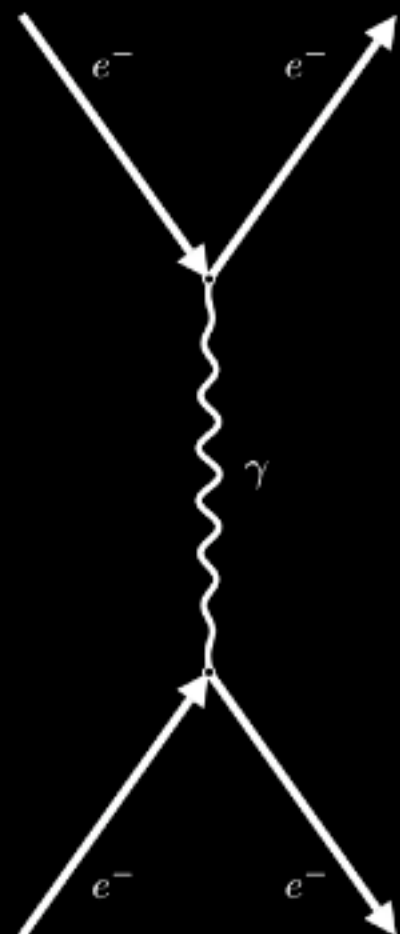
Charged-current weak scattering with W exchange



$$e^- e^- \rightarrow e^- e^-$$

QED scattering with virtual photon exchange

Distance

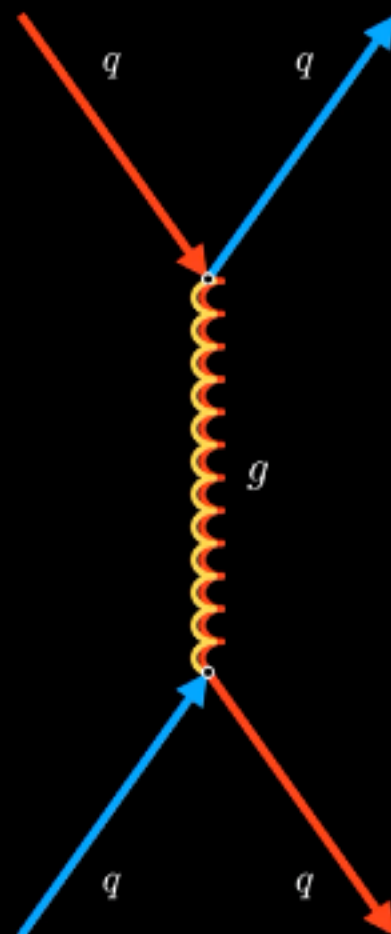


Time

$$q q \rightarrow q q$$

QCD scattering with virtual gluon exchange

Distance

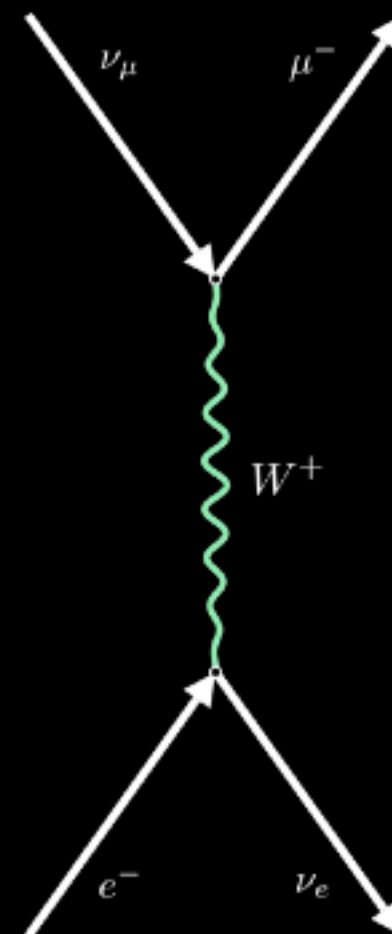


Time

$$\nu_\mu e^- \rightarrow \nu_e \mu^-$$

Charged-current weak scattering with W exchange

Distance



Time

Interactions

Carried by bosons (spin 1)